

A counterexample to Conjecture 00000008432

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The first assertion says that the hypergraph formed by the supports of minimum-weight codewords has rank $t + 1$ when it is a t -design. Under the standard definition, hypergraph rank is the maximum cardinality of an edge. The binary simplex code of length seven supplies a counterexample: its minimum-weight supports form a 2 -($7, 4, 2$) design, while their hypergraph rank is 4, not $2 + 1 = 3$.

The code and all its minimum supports. Label the seven coordinates by the nonzero vectors $v \in \mathbb{F}_2^3$, and define

$$c(u) = (u \cdot v)_{v \in \mathbb{F}_2^3 \setminus \{0\}}, \quad C = \{c(u) : u \in \mathbb{F}_2^3\}.$$

The dot product is over $\mathbb{F}_2$. The map c is linear and injective: the coordinates at the three standard basis vectors recover the three entries of u . Thus C is a binary linear code of dimension three. For $u \neq 0$, precisely four vectors $v \in \mathbb{F}_2^3$ satisfy $u \cdot v = 1$. None of them is zero. All seven nonzero codewords therefore have weight four; these are exactly the minimum-weight codewords. Their seven supports are distinct, since binary codewords are determined by their supports.

For any distinct nonzero coordinates v, w , these vectors are linearly independent over $\mathbb{F}_2$. The simultaneous equations $u \cdot v = u \cdot w = 1$ have exactly two solutions u , both nonzero. Hence every pair of coordinates lies in exactly two supports, proving the 2 -($7, 4, 2$) design property. Its strength is not three: if $z = v + w$, the three equations with right side 1 are inconsistent, whereas for three independent coordinate vectors they have exactly one solution. Thus $t = 2$ is the actual design strength, and not an arbitrarily chosen weaker parameter. Since every support has cardinality four, the hypergraph rank is exactly four. This disproves the rank assertion and therefore the full conjunction in the conjecture.

Formal verification. Lean represents binary triples by `Fin(8)` and all length-seven binary words by `Fin(128)`. The bit encoding is proved surjective onto every Boolean coordinate function and every predicate on the seven coordinates, and injective. The addition and scalar operations are verified to be the coordinate operations of $\mathbb{F}_2$. The encoder is the displayed dot product with all seven nonzero columns. Its linearity and injectivity, all codeword weights, all minimum supports, and the absence of repeated blocks are proved.

The pair-count theorem quantifies over every pair of distinct coordinates; the two explicit triple counts also certify that the strength is not three. The predicate `HypergraphRank` requires both a bound for every minimum support and a support attaining that bound. The final theorem `conjecture8432_counterexample` includes the linear-code and design hypotheses and proves rank four and failure of rank three. These facts are kernel-checked finite statements, not an external search certificate.

The proof uses Lean 4.19.0 with `Std` only and contains no admitted lemmas, native evaluation proofs, or custom axioms.